

Discrete Wave Mechanics: Deriving the Schrödinger Equation and the Mass Limit of Quantum Superposition from Vacuum Lattice Sintering

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Abstract

Standard Quantum Mechanics treats the complex wavefunction and its first-order time evolution as fundamental postulates. Expanding on the Selection-Stitch Model (SSM), we propose that these properties are emergent consequences of a discrete, crystallized vacuum. We define a “particle” as a stable, macroscopic topological braid (a Trefoil knot) propagating within a $K = 12$ Face-Centered Cubic (FCC) lattice. We model this vacuum as a Chiral Micropolar Continuum, where nodes possess both translational and rotational degrees of freedom. We explicitly derive the isotropic Laplacian from the 12 nearest-neighbor forces and introduce a novel Chiral Velocity Coupling arising from the Berry connection of the defect’s topology. We demonstrate that this coupling naturally generates the complex unit i , the global $U(1)$ symmetry, and the exact Schrödinger equation in the non-relativistic limit. Finally, using the geometrically renormalized lattice spacing ($a \approx 0.77l_P$) and the strict $a/\sqrt{3}$ kinematic exclusion limit of the vacuum, we predict a specific two-step mass limit for quantum coherence: a soft decoherence onset at $m_{soft} \approx 28\mu g$, and an absolute hard cutoff at $m_{hard} = \sqrt{3}m_{soft} \approx 49\mu g$. This distinguishing prediction provides a sharp, falsifiable geometric signature for the transition from quantum superposition to classical gravity.

1 Introduction

The duality of matter remains the central mystery of quantum foundations. While the Schrödinger equation predicts experimental outcomes with precision, it lacks a mechanical substrate. Approaches by ’t Hooft [1] and Volovik [2] suggest that quantum behavior emerges from deterministic discrete dynamics. The Selection-Stitch Model (SSM) operationalizes this by modeling the vacuum as a saturated $K = 12$ elastic lattice bounded by a strict internal kinematic exclusion limit [4]. We show that when the vacuum is treated as a Chiral Cosserat Continuum, the classical lattice dynamics naturally generate the complex wave equation.

2 Geometric Framework

We adopt the structural foundation of the SSM, which identifies the Face-Centered Cubic (FCC) lattice (Cuboctahedron coordination) as the mechanically stable ground state of the vacuum.

2.1 Cosserat Mechanics and Chirality

While the lattice geometry is fixed, the mechanics of the nodes determine the emergent physics. Standard elasticity treats nodes as point masses, yielding real-valued phonons. To recover Quantum Mechanics, we model the vacuum as a Micropolar (Cosserat) Continuum, where nodes possess both translational (u) and rotational (θ) degrees of freedom.

- **Spin:** The rotational mode θ provides the physical basis for particle spin and rest mass.
- **Chirality:** The particle defect is a macroscopic topological braid (a Trefoil knot, 3_1). This introduces Chirality into the local lattice deformations, creating the cross-coupling that generates the complex unit i .

3 Derivation from the Lattice Lagrangian

3.1 Discrete-to-Continuum Limit (The Isotropic Laplacian)

The Lagrangian formulation requires a continuous spatial derivative operator. We explicitly derive this from the discrete FCC lattice to justify the use of the isotropic Laplacian ∇^2 . The potential energy of a node at x is determined by the tension differences with its 12 nearest neighbors at vectors $\{e_i\}$. The FCC basis vectors are permutations of $\frac{a}{\sqrt{2}}(\pm 1, \pm 1, 0)$.

The lattice Laplacian is defined as the sum of first differences:

$$\Delta_{FCC}\psi(x) = \sum_{i=1}^{12} [\psi(x + e_i) - \psi(x)] \quad (1)$$

Performing a Taylor expansion:

$$\psi(x + e_i) \approx \psi(x) + (e_i \cdot \nabla)\psi + \frac{1}{2}(e_i \cdot \nabla)^2\psi \quad (2)$$

The linear terms vanish due to the inversion symmetry of the FCC lattice. The quadratic terms involve the sum $\sum (e_i \cdot \nabla)^2$. Explicit calculation shows that for the 12 FCC vectors, the diagonal terms sum to $8(\partial_x^2 + \partial_y^2 + \partial_z^2)$, while cross-terms cancel. Including the Taylor factor (1/2) and the lattice constant basis factor ($a^2/2$), we obtain:

$$\Delta_{FCC}\psi \approx \frac{1}{2} \cdot \frac{a^2}{2} \cdot 8\nabla^2\psi = 2a^2\nabla^2\psi \quad (3)$$

This rigorous convergence justifies the use of ∇^2 in the continuum Lagrangian below.

3.2 The Chiral Cosserat Action

We define the physics of the defect via a Lattice Lagrangian Density $\mathcal{L}$. The first terms represent standard Micropolar Elasticity (kinetic energy and elastic potential ω_0). The final term is the novel topological contribution: a time-asymmetric coupling representing the defect's chirality (spin).

$$\mathcal{L} = \frac{1}{2}(\partial_t u)^2 + \frac{1}{2}(\partial_t \theta)^2 - \frac{c^2}{2}(\nabla u)^2 - \frac{c^2}{2}(\nabla \theta)^2 - \frac{\omega_0^2}{2}(u^2 + \theta^2) + \underbrace{\Omega(u\dot{\theta} - \theta\dot{u})}_{\text{Chiral Coupling}} \quad (4)$$

Here, Ω is the gyroscopic coupling strength determined by the braid winding number.

3.3 Microscopic Origin of the Chiral Coupling

The coupling term $\Omega(u\dot{\theta} - \theta\dot{u})$ is not an ad hoc addition but arises fundamentally from the Berry connection of the discrete lattice wavefunction. As a topological braid moves through the lattice, the local basis vectors undergo a rotation $\mathcal{R}(t)$. The discrete hopping matrices Γ_j acquire a geometric phase factor $e^{i\gamma}$. In the continuum limit, this phase manifests as a gauge field A_μ coupled to the velocity:

$$A_0 \propto \langle \psi | \partial_t | \psi \rangle \approx u\dot{\theta} - \theta\dot{u} \quad (5)$$

Thus, the term representing "Chirality" in the macroscopic Lagrangian is the direct imprint of the Berry curvature generated by the non-trivial topology of the Trefoil knot defect.

3.4 Topological Origin of Parameters

The Lagrangian introduces two frequency scales: the elastic stiffness ω_0 and the chiral frequency Ω . In the SSM, these correspond to the two distinct topological features of the defect:

- **ω_0 (Lattice Stiffness):** Corresponds to the tension of the $K = 12$ bulk lattice. This represents the high-energy hadronic scale (Proton mass equivalent).
- **Ω (Gyroscopic Frequency):** Corresponds to the winding frequency of the braid's boundary membrane. This represents the low-energy leptonic scale (Electron mass equivalent).

The mass of the particle is derived as $m \propto \sqrt{\omega_0^2 + \Omega^2}$. If we identify the bulk stiffness with the proton mass scale and the braid frequency with the electron mass scale, the model naturally accommodates the hierarchy of particle masses $m_p/m_e \approx 1836$ as a ratio of bulk-to-defect coupling strengths [6].

3.5 Explicit Euler-Lagrange Derivation

The Euler-Lagrange equation is given by:

$$\partial_t \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right) + \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial (\nabla \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0 \quad (6)$$

We apply this independently for the fields u and θ .

For the radial field u :

$$\frac{\partial \mathcal{L}}{\partial \dot{u}} = \dot{u} - \Omega \theta, \quad \frac{\partial}{\partial t}(\dots) = \ddot{u} - \Omega \dot{\theta} \quad (7)$$

$$\frac{\partial \mathcal{L}}{\partial(\nabla u)} = -c^2 \nabla u, \quad \nabla \cdot (\dots) = -c^2 \nabla^2 u \quad (8)$$

$$\frac{\partial \mathcal{L}}{\partial u} = -\omega_0^2 u + \Omega \dot{\theta} \quad (9)$$

Combining terms:

$$(\ddot{u} - \Omega \dot{\theta}) + (-c^2 \nabla^2 u) - (-\omega_0^2 u + \Omega \dot{\theta}) = 0 \quad (10)$$

$$\Rightarrow \ddot{u} - 2\Omega \dot{\theta} - c^2 \nabla^2 u + \omega_0^2 u = 0 \quad (11)$$

For the torsional field θ :

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = \dot{\theta} + \Omega u, \quad \frac{\partial}{\partial t}(\dots) = \ddot{\theta} + \Omega \dot{u} \quad (12)$$

$$\frac{\partial \mathcal{L}}{\partial(\nabla \theta)} = -c^2 \nabla \theta, \quad \nabla \cdot (\dots) = -c^2 \nabla^2 \theta \quad (13)$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = -\omega_0^2 \theta - \Omega \dot{u} \quad (14)$$

Combining terms:

$$(\ddot{\theta} + \Omega \dot{u}) + (-c^2 \nabla^2 \theta) - (-\omega_0^2 \theta - \Omega \dot{u}) = 0 \quad (15)$$

$$\Rightarrow \ddot{\theta} + 2\Omega \dot{u} - c^2 \nabla^2 \theta + \omega_0^2 \theta = 0 \quad (16)$$

3.6 Complexification and Geometry

We define the complex field $\psi = u + i\theta$. Multiplying Eq. (16) by i and adding it to Eq. (11):

$$(\ddot{u} + i\ddot{\theta}) + 2\Omega(-\dot{\theta} + i\dot{u}) - c^2 \nabla^2(u + i\theta) + \omega_0^2(u + i\theta) = 0 \quad (17)$$

We rewrite the coupling term using the identity $-\dot{\theta} + i\dot{u} = i(\dot{u} + i\dot{\theta}) = i\dot{\psi}$. Substituting this back, we obtain the complex wave equation:

$$\frac{\partial^2 \psi}{\partial t^2} + 2i\Omega \frac{\partial \psi}{\partial t} - c^2 \nabla^2 \psi + \omega_0^2 \psi = 0 \quad (18)$$

This derivation proves that the complex unit i is the geometric operator mapping the chiral rotation between radial and torsional modes.

3.7 Unitarity and Probability Conservation

The Lagrangian possesses a global $U(1)$ symmetry. We rewrite the interaction term in complex variables as $-\frac{i\Omega}{2}(\psi^* \dot{\psi} - \dot{\psi}^* \psi)$. The associated conserved Noether charge density j^0 is:

$$j^0 = \text{Im}(\psi^* \dot{\psi}) + \Omega |\psi|^2 \quad (19)$$

As in standard Klein-Gordon theory, this relativistic charge is not strictly positive-definite for all modes. However, the physical probability interpretation emerges in the non-relativistic limit. As derived in Section IV, the dynamics reduce to the Schrödinger equation, where the conserved density is strictly $|\phi|^2$, guaranteeing positive-definite probability conservation for stable matter.

4 Evolution and the Schrödinger Limit

To recover standard mechanics, we transform to the Larmor Frame (co-rotating with the lattice precession) using the ansatz $\psi(x, t) = \Phi(x, t)e^{-i\Omega t}$. We calculate the time derivatives:

$$\dot{\psi} = (\dot{\Phi} - i\Omega\Phi)e^{-i\Omega t} \quad (20)$$

$$\ddot{\psi} = (\ddot{\Phi} - 2i\Omega\dot{\Phi} - \Omega^2\Phi)e^{-i\Omega t} \quad (21)$$

Substituting these into the complex wave equation:

$$(\ddot{\Phi} - 2i\Omega\dot{\Phi} - \Omega^2\Phi) + 2i\Omega(\dot{\Phi} - i\Omega\Phi) - c^2\nabla^2\Phi + \omega_0^2\Phi = 0 \quad (22)$$

The cross terms $-2i\Omega\dot{\Phi}$ and $+2i\Omega\dot{\Phi}$ cancel exactly. The terms $-\Omega^2\Phi$ and $+\Omega^2\Phi$ combine to $+\Omega^2\Phi$. The result is the exact Klein-Gordon equation:

$$\frac{\partial^2\Phi}{\partial t^2} - c^2\nabla^2\Phi + (\omega_0^2 + \Omega^2)\Phi = 0 \quad (23)$$

We identify the effective rest mass m via:

$$\frac{m^2c^4}{\hbar^2} = \omega_0^2 + \Omega^2 \Rightarrow m = \frac{\hbar}{c^2}\sqrt{\omega_0^2 + \Omega^2} \quad (24)$$

Finally, applying the non-relativistic envelope approximation ($\Phi(x, t) = \phi(x, t)e^{-imc^2t/\hbar}$ and neglecting $\dot{\phi}$), we recover the free Schrödinger equation:

$$-\frac{2im}{\hbar}\dot{\phi} - \nabla^2\phi = 0 \Rightarrow i\hbar\frac{\partial\phi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\phi \quad (25)$$

5 Discussion and Prediction

5.1 Limitations of Current Derivation

This derivation establishes a rigorously defined path from a chiral vacuum lattice to single-particle quantum mechanics. We acknowledge the following boundaries of the current scope:

- **Lagrangian Origin:** While motivated by the Berry connection of the Trefoil topological braid, the coefficients in Eq. (4) require further derivation from first-principles inter-atomic potentials.
- **Entanglement:** The lattice naturally supports non-local correlations through the shared elastic medium (the bulk connects all defects), but a rigorous derivation of Bell-type correlations remains an open challenge.

5.2 Derivation of the Two-Step Geometric Decoherence Limit

We now derive the specific mass scale for objective wave collapse based on the exact resolution and structural limits of the vacuum lattice. The reduced Compton wavelength λ_c defines the quantum resolution scale of a particle.

1. The Soft Limit (Schrödinger Breakdown): For a wavefunction to be consistently represented as a linear oscillation on the discrete lattice, its wavelength must exceed the macroscopic resting lattice spacing (a). Using the geometrically renormalized lattice spacing $a \approx 0.77l_P$ derived in our previous work [7], the soft limit is:

$$\lambda_c = \frac{\hbar}{mc} \geq a \quad \Rightarrow \quad m_{soft} = \frac{\hbar}{ac} \approx 28\mu g \quad (26)$$

At $\approx 28\mu g$, the continuous Schrödinger equation breaks down. The wavepacket begins to “feel” the discrete nodes, causing $\mathcal{O}(1)$ lattice corrections to dominate and lattice anisotropy to modify the dispersion relation.

2. The Hard Limit (The Metric Wall Cutoff): However, complete and absolute decoherence does not occur until the mass forces the local vacuum nodes against the fundamental metric wall of the network [4]. The absolute kinematic exclusion radius of the unitary stitch is $a/\sqrt{3}$. When the Compton wavelength compresses to this hard geometric boundary, the lattice structurally shatters if forced to compress further. This establishes the strict metric cutoff:

$$\lambda_c = \frac{\hbar}{mc} \geq \frac{a}{\sqrt{3}} \quad \Rightarrow \quad m_{hard} = \frac{\hbar}{(a/\sqrt{3})c} = \sqrt{3}m_{soft} \approx 49\mu g \quad (27)$$

The Gray Zone: Between $28\mu g$ and $49\mu g$, the model predicts a specific “gray zone” of macroscopic reality. Superposition degrades exponentially as structural lattice tension rises, but the lattice remains unbroken. At exactly $49\mu g$, the wave “bottoms out” against the metric wall. The medium physically cannot stretch any further to support linear wave oscillation, forcing instantaneous collapse into a rigid, classical geometric deformation.

5.3 Comparison with Penrose Objective Reduction

The SSM prediction of a two-step decoherence limit provides a strictly falsifiable, stair-step signature that separates it from standard continuous objective reduction models. The Diósi-Penrose model [8] posits that quantum state reduction is a gravitational phenomenon occurring near the Planck Mass ($\sim 21.7\mu g$) due to a fundamental, abstract conflict between General Relativity and Quantum Linearity. Penrose predicts a single, fuzzy instability threshold.

In contrast, the SSM identifies a distinct, mechanical sequence: an onset of severe lattice corrections at $28\mu g$ followed by an absolute structural cutoff at $49\mu g$. The convergence of these relativistic and geometric approaches upon nearly the exact same magnitude ($\sim 10^{-5}$ g) strongly suggests this is a fundamental physical boundary. Next-generation macroscopic superposition experiments (such as MAQRO) possessing sufficient resolution could actively hunt for this characteristic two-step degradation, providing direct experimental evidence for the discrete lattice.

6 Conclusion

The Selection-Stitch Model provides a rigorous derivation of Quantum Mechanics from the mechanics of a Chiral Cosserat Vacuum. By identifying the origin of the complex unit, probability conservation, and the Schrödinger equation in lattice gyroscopics governed by topological braids, we offer a testable geometric foundation for physical reality.

Furthermore, by framing mass as lattice tension, we identify the exact mechanical cause of objective wave collapse: a two-step process initiated by the macroscopic lattice spacing ($a \approx 0.77l_P$) and terminally halted by the absolute $a/\sqrt{3}$ metric wall.

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